

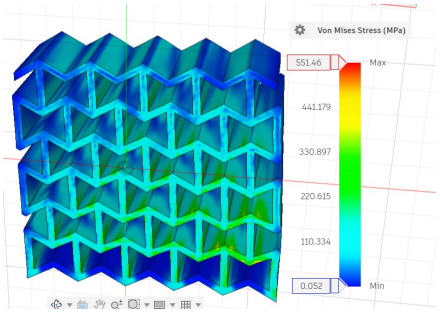
Midsole Lattice Optimisation

Core objective: Parametric comparison of three TPU lattice geometries under an 80 kg runner heel-strike load (2,000 N) to maximise Specific Energy Absorption (SEA) while reducing weight.

Table of key information

	Solid Baseline block	Re-entrant Auxetic (bowtie)	Hexagonal Honeycomb	Diamond Grid
TPU Mass (g)	60	23.97	23.31	23.3
Mass reduction (%)	0	60.1	61.2	61.2
Comparison of how well the designs perform against each other:				
Peak stress (MPa)	-	551.46	453.79	394.99
Deformation in direction of load (mm)	-	772.99	514.20	509.44
Stiffness (N/mm)	-	2.59	3.89	3.93
SEA (J/kg)	-	32,248	22,059	21,864

- **Performance Winner**: Re-entrant Auxetic gives the highest SEA (32,248 J/kg, about 46% above the others) and the most compliance (2.59 N/mm), ideal for heel cushioning. Diamond Grid is the stiffest (3.93 N/mm) with the lowest peak stress (394.99 MPa, 28% lower).
- **FEA Singularity & Stress Concentration**: Auxetic peaks at 551.46 MPa at un-filleted re-entrant joints (a linear-FEA singularity); add R = 0.5 mm fillets to mitigate fatigue.



- **Mesh Independence**: A 1.5 mm element baseline was validated against 3.0 mm and 1.0 mm meshes to balance solve speed and bending resolution.

Design for Additive Manufacturing Validation

- **Printability**: 90°draft angles eliminate internal supports entirely. Uniform 1.5 mmwalls map directly to 3–4 perimeter passes on a 0.4 mmnozzle.
- **Slicing Validation**: Kiri:Moto slicer audits confirm print times of 4h 45m to 5h 00m utilizing ≈ 26g of TPU filament per unit.

Scan or click here for raw CAD files, FEA simulation datasets, and engineering drawing PDFs

METHODOLOGICAL NOTE: LINEAR FEA SCOPE & DISPLACEMENT INTERPRETATION
Linear solver: displacements above 50 mm come from linear static assumptions (F = Kx). **Excluded factors:** TPU hyperelastic limits, geometric nonlinearity (NLGEOM) and cell-wall contact/crushing are not modelled. **Purpose:** a quick baseline to rank stiffness (K) and cushioning across geometries, not absolute values.